

Erdős Problem #142

EP-142

L1 - TRACTABLE

OPEN

Combinatorics

ORIGINAL PROBLEM STATEMENT

Let $r_k(N)$ be the largest possible size of a subset of $\{1, \dots, N\}$ that does not contain any non-trivial k -term arithmetic progression. Prove an asymptotic formula for $r_k(N)$.

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>